

Siheng Xia

Undergraduate Student in Tsinghua University

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Education

Tsinghua University

B.S. in Mathematics and Physics & B.Eng. in Software Engineering

Aug. 2024 – June 2028 (Expected)

Beijing, China

- GPA: 4.0/4.0 | Ranking: **1/49**
- Selected Coursework: Advanced Calculus (A+), Data Structures and Algorithms (A+), Linear Algebra (A), Probability Theory (A), Statistical Inference (A), Machine Learning (A-), Topology (A-)
- Research Interests: Robotics, Generative Models

Carnegie Mellon University

Exchange Student at School of Computer Science

Aug. 2026 – Dec. 2026 (Expected)

Pittsburgh, PA, USA

Publications

[C1] Tianming Wei, Yufeng Tian, Jinxian Zhou, **Siheng Xia**, Zhecheng Yuan, and Huazhe Xu.
[DexReprise: Reference-Guided Generalist Dexterous Manipulation from Human Motion and Videos.](#)
CoRL 2026 Conference Submission (under review)

Research Experience

Tsinghua Embodied AI Lab (TEA Lab), Tsinghua University

Research Intern

Oct. 2025 – May 2026

Beijing, China

- **Advisor:** Prof. Huazhe Xu (Institute for Interdisciplinary Information Sciences)
- **DexReprise:** Built a reference-guided framework that converts offline human demonstrations (mocap & videos) into closed-loop dexterous robot policies, removing the need for real-time teleoperation.
- **Core Contribution (Algorithmic):** Refined the distillation pipeline by designing a dual-objective spatial-awareness auxiliary task. Formulated an observation loss for latent 3D coordinate retention and a relative delta loss for geometric discrepancy modeling.
- **Role & outcome:** Sole sophomore on a team of seniors and PhD students; contributed across the full research cycle. Co-authored *DexReprise* (CoRL 2026, under review).

LIU Lab, Tsinghua University

Research Intern

May 2026 – Present

Beijing, China

- **Advisor:** Prof. Ziming Liu (College of AI)
- **Belief-factorized world models:** Investigated whether a world model's responsibilities – inferring hidden state, maintaining belief over what is currently unobserved, and evolving physics forward in time – should be handled by separate modules rather than learned jointly; built toy-physics testbeds (Kepler orbits, occluded billiards) to test this factorization under occlusion.

Honors & Awards

2026	Meritorious Student Awarded for academic and leadership excellence	<i>Tsinghua University</i>
2025	National Scholarship Highest national scholarship for undergraduate students in China	<i>Ministry of Education, China</i>
2025	First-Class Comprehensive Excellence Scholarship Top comprehensive scholarship awarded for academic and leadership excellence	<i>Tsinghua University</i>
2025	Social Work Excellence Scholarship Awarded for outstanding contributions to student leadership and social service	<i>Tsinghua University</i>
2023	Silver Medal, Chinese Mathematical Olympiad (CMO) Ranked 270th among 650+ national finalists	<i>Chinese Mathematical Society</i>
2023	Gold Medal, Chinese Southeastern Mathematical Olympiad (CSMO) Ranked 8th among 1700+ participants nationwide	<i>Chinese Mathematical Society</i>
2022	Excellent Performance, Tsinghua YAU Summer Mathcamp for Secondary School Students Selected national program for mathematically talented high school students	<i>Tsinghua University</i>
2021 – 2023	First Prize, National Senior High School Mathematical Competition Qualification competition for the Chinese Mathematical Olympiad (CMO)	<i>Chinese Mathematical Society</i>

Teaching Experience

Object Oriented Programming <i>Undergraduate Teaching Assistant</i>	Spring 2026 <i>Tsinghua University</i>
• Instructor: Prof. Zhiyuan Liu (Department of Computer Science and Technology) • Mentored students on OOP and assignments; designed original homework problems.	

Technical Skills

- **Programming Languages:** Python, C++
- **ML / Robotics:** PyTorch, Isaac Sim, Mujoco, CuRobo
- **Tools & Infrastructure:** Git, Linux/Bash, CUDA
- **Languages:** English (Fluent, IELTS: 7.5/9), Chinese (Native)

Leadership and Service

• Deputy Head , Organization Department, Student Committee, Weiyang College	Nov 2025 – Present
• Secretary , Class Committee (Weiyang 40), Weiyang College	Sep 2024 – Sep 2025
• Captain , Ultimate Frisbee Team, Weiyang College	Nov 2025 – Present